BOILEAU GUILLAUME

Ingénieur Moyens d'Essais | Bancs de test, acquisition, LabVIEW, signal & validation

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EXPERIENCE

Software Engineer – System Integration, Hardware Interfaces & Operational Diagnostics

VEV Platform Services France

2025 – 2026 Valbonne, France

Context: Development and integration of software services for an EV charging CPMS platform connected to field equipment, operational data flows and hardware infrastructure.
Objective: Support reliable software integration, operational data-flow continuity, interface diagnostics and technical follow-up for services connected to charging station systems.

- **Software development:** Developed and maintained web and backend services using TypeScript, Angular and Node.js in an operational industrial environment.
- **System integration:** Worked at the interface between software services, field equipment constraints, operational data and production expectations.
- **Hardware/software interfaces:** Analysed interactions between platform services, connected charging stations, data exchanges and operational behaviours.
- **Operational data acquisition:** Monitored and analysed field-related data flows, including FTP-based exchanges used for technical follow-up.
- **Diagnostics:** Investigated incidents involving software behaviour, data consistency, interface continuity and equipment-related operational effects.
- **Validation logic:** Checked coherence between data flows, application behaviour, equipment states and expected operational results.
- **Monitoring & tooling:** Used Git, Zenhub, Confluence, OpenSearch and Sentry for issue tracking, diagnostic follow-up and technical documentation.

Senior R&D Engineer – Test Logic, Signal Processing & Validation Workflows

CNES, Laboratoire Lagrange, Observatoire de la Côte d'Azur

2023 – 2025 Nice, France

Context: Engineering activities for the LISA ESA/NASA space mission, involving complex software chains, simulation outputs, signal analysis, performance assessment and validation campaigns.
Objective: Develop, integrate and validate Python-based workflows for model comparison, signal/data analysis, consistency checks and traceable technical review.

- **Validation workflows:** Structured comparison, verification and validation logic for heterogeneous simulation outputs and technical deliverables.
- **Signal processing:** Analysed weak, noisy and superposed signals using spectral methods, signal/noise separation logic and performance metrics.
- **Measurement-oriented analysis:** Worked on simulated and instrument-related data where performance depends on noise, sensitivity, acquisition quality and signal interpretation.
- **Automated test logic:** Defined repeatable analysis sequences, comparison rules, consistency checks and acceptance-oriented diagnostics.
- **Data reliability:** Ensured traceability of inputs, configurations, outputs, assumptions, metadata and validation results.
- **Software platform:** Developed CATpyLISA, a Python platform for model integration, data comparison, validation and technical review. GitLab:CATpyLISA
- **Technical documentation:** Used Confluence, GitLab, Jupyter and LaTeX to document methods, validation results and technical evidence.
- **Coordination:** Coordinated technical contributions, supervised interns and supported collaborative validation activities in an international mission environment.

Data Scientist – Noise Diagnostics, Measurement Analysis & Performance Validation

Universiteit Antwerpen

2021 – 2023 Antwerp, Belgium

Context: Data analysis and performance studies for precision instruments, with focus on noisy measurements, environmental disturbances and sensitivity limitations.
Objective: Identify, characterize and mitigate sources affecting measurement quality and system sensitivity using reproducible statistical and signal-processing workflows.

- **Test data analysis:** Analysed measurement and simulation outputs to identify abnormal behaviours, noise contributions and degradation sources.
- **Signal & noise diagnostics:** Developed methods for spectral analysis, noise characterization, sensitivity studies and cross-sensor correlations.
- **Python pipelines:** Built reproducible analysis workflows for data cleaning, transformation, modelling, diagnostics and reporting.
- **Performance validation:** Assessed system sensitivity, measurement stability and limitations through statistical and signal-based criteria.
- **Technical reporting:** Produced traceable analyses and technical summaries for international engineering and research stakeholders.

PROFILE FOR TERX

Engineer at the interface between test means, software development, signal processing, acquisition-oriented data analysis and validation of complex systems. Background developed in space, precision instrumentation and industrial software environments, with strong experience in measurement-quality analysis, automated workflows, traceability and technical diagnostics. Relevant for a role in **test bench software, instrumentation, acquisition, automated test sequences and performance validation**: able to understand physical measurements, structure robust analysis logic, develop technical software, investigate anomalies, document results and work with multidisciplinary teams.

FIT FOR TEST MEANS

- Strong match with **signal processing**, measurement analysis, noise characterization and performance assessment.
- Experience building **automated analysis workflows** and repeatable validation sequences in Python.
- Familiarity with **LabVIEW** from academic training; ready to reactivate and deepen it in an industrial test-bench context.
- Good understanding of **instrumentation logic**: acquisition, signal quality, data reliability, calibration-oriented reasoning, diagnostics and exploitation of measurements.
- Experience with **software/hardware interfaces** through operational systems connected to field equipment.
- Strong practice of **technical documentation**, result traceability, validation reports and collaborative engineering tools.
- Comfortable in R&D, production-support and multidisciplinary environments involving software, data, instrumentation and system constraints.

TECHNICAL SKILLS

Python · LabVIEW · Test logic · Signal processing · Data acquisition · Measurement analysis · Noise analysis · Spectral analysis · Automation · Validation · IVVQ · V-cycle · Test benches · Instrumentation · Hardware/software interfaces · Electrical test support · EV charging stations · Communication buses · Soldering · Maintenance · B0 / BE Essais · Data reliability · Traceability · Technical reporting · Linux · Git · GitLab · Docker · Confluence · Zenhub · OpenSearch · Sentry · Jupyter · SQL · C/C++ · Matlab · TypeScript · Angular · Node.js

METHODS & ENGINEERING

- Test sequences, test logic, integration checks, verification and validation criteria.
- Acquisition/data-flow monitoring, consistency checks, anomaly diagnostics and root-cause-oriented analysis.
- Signal/noise analysis, SNR reasoning, performance validation and robustness assessment.
- Technical documentation, validation evidence, reporting and traceability of assumptions/results.
- Agile delivery, task tracking, technical coordination and collaborative engineering.

Junior R&D Engineer – Instrumental Data, Simulation & Signal Analysis

ARTEMIS, Observatoire de la Côte d'Azur

2018 – 2021 | Nice, France

Context: R&D activities linked to weak-signal analysis, numerical simulation, instrumental diagnostics and gravitational-wave mission preparation.
Objective: Develop analysis methods and simulation workflows for complex, noisy and large-scale datasets.

- **Signal analysis:** Developed methods for the analysis and separation of weak, overlapping signals in noisy environments.
- **Simulation workflows:** Built numerical workflows for repeated analysis campaigns, uncertainty studies and performance assessment.
- **Instrumental diagnostics:** Investigated noise sources and limiting factors using correlation analysis and signal-processing methods.
- **Validation evidence:** Compared models, simulations and outputs to assess consistency, robustness and technical limitations.

LANGUAGES

English ●●●●○

Spanish ●●○○○

EDUCATION

PhD in Physics

Université Côte d'Azur

2018 – 2021 | Nice, France

MSc in Astronomy, Astrophysics & Space Engineering

Observatoire de Paris / Université Paris-Saclay

2017 – 2018 | Paris, France

Selective Graduate Program in Fundamental Physics

Université Paris-Saclay

2016 – 2018 | Orsay, France

BSc in Fundamental Physics

Université Paris-Saclay

2015 – 2016 | Orsay, France

PROFESSIONAL TRAINING

Supervised Machine Learning

DeepLearning.AI – Andrew Ng

2020 | Coursera

Recherche reproductible : principes méthodologi pour une science transparente

FUN MOOC – Inria

2025 | France Université Numérique

Continuous Professional Training – Software, Data, AI and Systems

OpenClassrooms

2024 – 2026

Python, SQL, Machine Learning, AI, Linux, JavaScript, TypeScript, Angular, Node.js, Django, REST APIs, C/C++, Java, algorithms and software engineering fundamentals.

PROJECTS

CNES / LISA – Validation, Signal Processing & Automated Analysis Workflows

ESA/NASA Space Mission – Large-scale International Collaboration

2023 – 2025 | Nice, France

Contribution to software, signal-processing and validation workflows for the LISA space mission, with emphasis on traceable comparison of outputs, noisy signal analysis, performance diagnostics and consolidation of technical evidence.
Relevance for test means: automated validation logic, comparison sequences, signal/noise analysis, consistency checks, traceable outputs, technical documentation and multidisciplinary coordination.

Precision Instrumentation – Noise Diagnostics & Sensitivity Studies

Universiteit Antwerpen / Gravitational-wave Instrumentation Context

2021 – 2023 | Antwerp, Belgium

Development of statistical and signal-processing workflows to identify noise sources, evaluate measurement quality and support performance improvement for precision instruments.
Relevance for instrumentation: measurement analysis, spectral diagnostics, cross-sensor correlations, data reliability, technical reporting and performance validation.

Operational Industrial Platform – Hardware-connected Software Services

VEV Platform Services France

2025 – 2026 Valbonne, France

Software development and diagnostic support for a CPMS platform connected to EV charging infrastructure, involving operational data flows, production incidents and hardware/software interface analysis.
Relevance for industrial testing: interface diagnostics, data-flow monitoring, incident analysis, software reliability and operational traceability.